Type MS · Finding an unknown constant from a condition on the roots

Answer 1 of 10 · Question 5 part c · 3 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

5 a	$\alpha+\beta=-\frac{b}{a}=\frac{-(6+2p)}{2}=-3-p$	B1
	$\alpha\beta=\frac{c}{a}=\frac{2p}{2}=p$	B1 (2)
b	$(\alpha-\beta)^2=\alpha^2+\beta^2-2\alpha\beta$	M1
	$=(\alpha+\beta)^2-4\alpha\beta$	M1
	$=(-3-p)^2-4p$	M1
	$=9+2p+p^2$ *	A1 cso (4)
c	$(\alpha-\beta)=3$ implies $(\alpha-\beta)^2=9$	B1
	So $9=9+2p+p^2\Rightarrow 0=2p+p^2\Rightarrow 0=p(2+p)$	M1
	$p=0$ or $p=-2$	A1 (3)
		Total 9 marks

Part	Mark	Notes
(a)	**B1**	For the sum $\alpha+\beta=-3-p$ Need not be simplified. E.g., accept $\frac{-(6+2p)}{2}$
	B1	For the product $\alpha\beta=p$ $\left(\text{Accept } \frac{2p}{2}\right)$
(b)	**M1**	For the correct algebra on $(\alpha-\beta)^2$ $(\alpha-\beta)^2=\alpha^2+\beta^2-2\alpha\beta$ Accept un-simplified and terms in any order
	M1	For the **correct** algebra on $(\alpha-\beta)^2$ using $(\alpha+\beta)^2$ $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$ Note: $(\alpha+\beta)^2=9+6p+p^2$ which you may see substituted in without seeing the algebra
	M1	For substituting in their sum and product from part (a) $(\alpha-\beta)^2=(-3-p)^2-4p$
	A1 cso	For obtaining the given result $(\alpha-\beta)^2=9+2p+p^2$ Note: condone absence of seeing the LHS. **Note**: **This is a given result.** There must be **no errors or omissions** (including in algebra) in their work for this mark
(c)	**B1**	For $(\alpha-\beta)^2=9$
	M1	For setting the given answer = 9 $9+2p+p^2=9\Rightarrow 2p+p^2=0\Rightarrow p=\ldots$ Two values.
	A1	For $p=0$ or $p=-2$

Type MS · Finding an unknown constant from a condition on the roots

Answer 1 of 10 · Question 5 part c · 3 marks

2022 JUN · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Scheme	Marks
	$\cos 2A = \cos^2 A - \sin^2 A$	

Type MS · Finding an unknown constant from a condition on the roots

Answer 2 of 10 · Question 11 part a.ii · 5 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

11 (a)	(i) $2\alpha\beta = 3 \Rightarrow q = \alpha\beta \Rightarrow q = \frac{3}{2}$ (ii) $\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta$, $\left[\alpha^2 + \beta^2 = \frac{k^2-6k-3}{4}\right]$ $(\alpha+\beta)^2 = \frac{k^2-6k-3}{4} + 3 = \frac{k^2-6k+9}{4} = \frac{(k-3)^2}{4}$ $\Rightarrow \alpha+\beta = \frac{k-3}{2} \Rightarrow p = \frac{k-3}{2}$	B1 M1A1 M1A1 [5]
(b)	$7\left(\frac{3}{2}\right) = 3\left(\frac{k-3}{2}\right) \Rightarrow 7 = k-3 \Rightarrow k = 10$	M1A1 [2]
(c)	$\alpha+\beta = \frac{10-3}{2} = \frac{7}{2}$ Sum: $\frac{\alpha}{\alpha+\beta} + \frac{\beta}{\alpha+\beta} = \frac{\alpha+\beta}{\alpha+\beta} = 1$ Product: $\frac{\alpha}{\alpha+\beta} \times \frac{\beta}{\alpha+\beta} = \frac{\alpha\beta}{(\alpha+\beta)^2} = \frac{\frac{3}{2}}{\left(\frac{7}{2}\right)^2} = \frac{6}{49}$ Equation: $x^2 - x + \frac{6}{49}$ $(=0)$ $49x^2 - 49x + 6 = 0$	M1 B1 M1 M1 A1 [5]

(a) (i)	B1	$q = \frac{3}{2}$
(ii)	M1	State or use any correct arrangement of $(\alpha+\beta)^2 = \alpha^2 + \beta^2 + 2\alpha\beta$ and substitute $\alpha\beta = \frac{3}{2}$ and $\alpha^2 + \beta^2 = \frac{k^2-6k-3}{4}$ oe p may be seen instead of $\alpha+\beta$ at any stage of the working. Condone $-p = \alpha+\beta$ for this mark only.
	A1	Any correct expression for $(\alpha+\beta)^2$ or $4(\alpha+\beta)^2$ in terms of k
	M1	Correct method to find an expression for p in terms of k from their expression for $(\alpha+\beta)^2$ or $4(\alpha+\beta)^2$. Ignore negative square root. Must identify $p = \alpha+\beta$.
	A1	A correct expression for p in terms of k. The question does not demand simplification, but if it is left as a square root there must be an indication that p must be the positive root (ie no $\pm$ present).

Type MS · Finding an unknown constant from a condition on the roots

Answer 2 of 10 · Question 11 part a.ii · 5 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

(b)	M1	Substitute $\alpha\beta = \frac{3}{2}$ and their expression for $\alpha+\beta$ from (a)(ii) into $7\alpha\beta = 3(\alpha+\beta)$ and obtain an equation in k
	A1	$k = 10$ Do not accept if there is an additional root, such as $k = -4$.
(c)	M1	Find a value for $\alpha+\beta$. Need not be simplified. Follow through their k and their expression for p.
	B1	Show or state sum of roots = 1 (stated or explicitly used).
	M1	Write product of roots as $\frac{3/2}{(7/2)^2}$, ft their $\alpha+\beta = \frac{7}{2}$.
	M1	Use equation is $x^2 -$ (sum of roots) + (product of roots) $(=0)$ $(=0$ may be missing). There must be some attempt to substitute values, not just expressions in terms of α and β. May be implied by an equation with fractions correctly cleared.
	A1	$49x^2 - 49x + 6 = 0$ Correct equation with integer coefficients. Must have = 0.

Type MS · Finding an unknown constant from a condition on the roots

Answer 3 of 10 · Question 11 part b · 2 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

11 (a)	(i) $2\alpha\beta = 3 \Rightarrow q = \alpha\beta \Rightarrow q = \frac{3}{2}$ (ii) $\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta$, $\left[\alpha^2+\beta^2 = \frac{k^2-6k-3}{4}\right]$ $(\alpha+\beta)^2 = \frac{k^2-6k-3}{4} + 3 = \frac{k^2-6k+9}{4} = \frac{(k-3)^2}{4}$ $\Rightarrow \alpha+\beta = \frac{k-3}{2} \Rightarrow p = \frac{k-3}{2}$	B1 M1A1 M1A1 [5]
(b)	$7\left(\frac{3}{2}\right) = 3\left(\frac{k-3}{2}\right) \Rightarrow 7 = k-3 \Rightarrow k = 10$	M1A1 [2]
(c)	$\alpha+\beta = \frac{10-3}{2} = \frac{7}{2}$ Sum: $\frac{\alpha}{\alpha+\beta} + \frac{\beta}{\alpha+\beta} = \frac{\alpha+\beta}{\alpha+\beta} = 1$ Product: $\frac{\alpha}{\alpha+\beta} \times \frac{\beta}{\alpha+\beta} = \frac{\alpha\beta}{(\alpha+\beta)^2} = \frac{\frac{3}{2}}{\left(\frac{7}{2}\right)^2} = \frac{6}{49}$ Equation: $x^2 - x + \frac{6}{49}$ $(=0)$ $49x^2 - 49x + 6 = 0$	M1 B1 M1 M1 A1 [5]

(a) (i)	B1	$q = \frac{3}{2}$
(ii)	M1	State or use any correct arrangement of $(\alpha+\beta)^2 = \alpha^2+\beta^2+2\alpha\beta$ and substitute $\alpha\beta = \frac{3}{2}$ and $\alpha^2+\beta^2 = \frac{k^2-6k-3}{4}$ oe p may be seen instead of $\alpha+\beta$ at any stage of the working. Condone $-p = \alpha+\beta$ for this mark only.
	A1	Any correct expression for $(\alpha+\beta)^2$ or $4(\alpha+\beta)^2$ in terms of k
	M1	Correct method to find an expression for p in terms of k from their expression for $(\alpha+\beta)^2$ or $4(\alpha+\beta)^2$. Ignore negative square root. Must identify $p = \alpha+\beta$.
	A1	A correct expression for p in terms of k. The question does not demand simplification, but if it is left as a square root there must be an indication that p must be the positive root (ie no $\pm$ present).

Type MS · Finding an unknown constant from a condition on the roots

Answer 3 of 10 · Question 11 part b · 2 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

(b)	M1	Substitute $\alpha\beta = \frac{3}{2}$ and their expression for $\alpha+\beta$ from (a)(ii) into $7\alpha\beta = 3(\alpha+\beta)$ and obtain an equation in k
	A1	$k = 10$ Do not accept if there is an additional root, such as $k = -4$.
(c)	M1	Find a value for $\alpha+\beta$. Need not be simplified. Follow through their k and their expression for p.
	B1	Show or state sum of roots = 1 (stated or explicitly used).
	M1	Write product of roots as $\frac{3/2}{(7/2)^2}$, ft their $\alpha+\beta = \frac{7}{2}$.
	M1	Use equation is $x^2 -$ (sum of roots) + (product of roots) $(=0)$ $(=0$ may be missing). There must be some attempt to substitute values, not just expressions in terms of α and β. May be implied by an equation with fractions correctly cleared.
	A1	$49x^2 - 49x + 6 = 0$ Correct equation with integer coefficients. Must have = 0.

Type MS · Finding an unknown constant from a condition on the roots

Answer 4 of 10 · Question 8 part b · 5 marks

2023 JAN · Paper 02 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

8(a)	$\alpha+\beta=\frac{k}{3}$ and $\alpha\beta=-\frac{1}{3}$	B1
	$(\alpha+\beta)^2=\alpha^2+2\alpha\beta+\beta^2\Rightarrow\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta$	M1
	$\alpha^2+\beta^2=\left(\frac{k}{3}\right)^3-2\left(-\frac{1}{3}\right)=\frac{k^2}{9}+\frac{2}{3}=\frac{k^2+6}{9}*$	A1 cso [3]
(b)	$\alpha^4+\beta^4=(\alpha^2+\beta^2)^2-2\alpha^2\beta^2$	M1
	$\frac{466}{81}=\left(\frac{k^2+6}{9}\right)^2-2\left(-\frac{1}{3}\right)^2$	M1
	$\frac{466}{81}=\left(\frac{k^2+6}{9}\right)^2-2\left(-\frac{1}{3}\right)^2\Rightarrow k^4+12k^2-448=0$	M1
	$k^4+12k^2-448=(k^2-16)(k^2+28)=0\Rightarrow k^2=16\Rightarrow k=4$	M1A1 [5]
(c)	Sum: $\frac{\alpha^3+\beta}{\beta}+\frac{\beta^3+\alpha}{\alpha}=\frac{\alpha^4+\beta^4+2\alpha\beta}{\alpha\beta}$	M1
	$\frac{\alpha^3+\beta}{\beta}+\frac{\beta^3+\alpha}{\alpha}=\frac{\frac{466}{81}+2\left(-\frac{1}{3}\right)}{-\frac{1}{3}}=-\frac{412}{27}$	A1
	Product: $\left(\frac{\alpha^3+\beta}{\beta}\right)\left(\frac{\beta^3+\alpha}{\alpha}\right)=\frac{\alpha^4+\beta^4+(\alpha\beta)^3+\alpha\beta}{\alpha\beta}$	M1
	$\left(\frac{\alpha^3+\beta}{\beta}\right)\left(\frac{\beta^3+\alpha}{\alpha}\right)=\frac{\frac{466}{81}+\left(-\frac{1}{3}\right)^3+\left(-\frac{1}{3}\right)}{\left(-\frac{1}{3}\right)}=-\frac{436}{27}$	A1
	3TQ: $x^2+\frac{412}{27}x-\frac{436}{27}=0\Rightarrow 27x^2+412x-436=0$	M1A1 [6]
		Total 14 marks

Question	Scheme	Marks
8(a)	$\alpha+\beta=\frac{k}{3}$ and $\alpha\beta=-\frac{1}{3}$	B1
	For the correct algebra on $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta$	M1
	For substituting *their* sum and *their* product to give a simplified expression for $\alpha^2+\beta^2=\left(\frac{k}{3}\right)^3-2\left(-\frac{1}{3}\right)=\frac{k^2}{9}+\frac{2}{3}=\frac{k^2+6}{9}*$	A1 cso [3]

Type MS · Finding an unknown constant from a condition on the roots

Answer 4 of 10 · Question 8 part b · 5 marks

2023 JAN · Paper 02 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

(b)	For the correct algebra on $\alpha^4+\beta^4=\left(\alpha^2+\beta^2\right)^2-2\alpha^2\beta^2$ Or $\alpha^4+\beta^4=(\alpha+\beta)^4-4\alpha\beta(\alpha^2+\beta^2)-6\alpha^2\beta^2$	M1
	For substituting the values for $\alpha^2+\beta^2$ (in terms of k) and $\alpha\beta$ into their $\alpha^4+\beta^4$ and equating to $\frac{466}{81}$ $\frac{466}{81}=\left(\frac{k^2+6}{9}\right)^2-2\left(-\frac{1}{3}\right)^2$ Or for substituting the values for $\alpha^2+\beta^2$ (in terms of k), $\alpha+\beta$ and $\alpha\beta$ into their $\alpha^4+\beta^4$ and equating to $\frac{466}{81}$	M1
	For forming a simplified quadratic. $\frac{466}{81}=\left(\frac{k^2+6}{9}\right)^2-2\left(-\frac{1}{3}\right)^2\Rightarrow k^4+12k^2-448=0$ Or for $\left(\frac{k^2+6}{9}\right)^2=\frac{448}{81}$ Their quadratic must come from an attempt at $\alpha^4+\beta^4$	M1
	For solving their quadratic. $k^4+12k^2-448=\left(k^2-16\right)\left(k^2+28\right)=0\Rightarrow k^2=16\Rightarrow k=4$ Their quadratic must come from an attempt at $\alpha^4+\beta^4$	M1
	For the correct value of $k=4$ If they give $k=\pm4$ this is A0	A1 [5]
(c)	For the correct algebra on the sum. $\frac{\alpha^3+\beta}{\beta}+\frac{\beta^3+\alpha}{\alpha}=\frac{\alpha^4+\beta^4+2\alpha\beta}{\alpha\beta}$	M1
	For the correct value of the sum. $\frac{\alpha^3+\beta}{\beta}+\frac{\beta^3+\alpha}{\alpha}=\frac{\frac{466}{81}+2\left(-\frac{1}{3}\right)}{-\frac{1}{3}}=-\frac{412}{27}$	A1
	For the correct algebra on the product. $\left(\frac{\alpha^3+\beta}{\beta}\right)\left(\frac{\beta^3+\alpha}{\alpha}\right)=\frac{\alpha^4+\beta^4+(\alpha\beta)^3+\alpha\beta}{\alpha\beta}$	M1
	For the correct value of the product. $\left(\frac{\alpha^3+\beta}{\beta}\right)\left(\frac{\beta^3+\alpha}{\alpha}\right)=\frac{\frac{466}{81}+\left(-\frac{1}{3}\right)^3+\left(-\frac{1}{3}\right)}{\left(-\frac{1}{3}\right)}=-\frac{436}{27}$	A1
	For the 3TQ from their values of the sum and product (used correctly) [= 0 not required for this mark].	M1

Type MS · Finding an unknown constant from a condition on the roots

Answer 4 of 10 · Question 8 part b · 5 marks

2023 JAN · Paper 02 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

	$x^2+\frac{412}{27}x-\frac{436}{27}=(0)$	
	For the correct 3TQ. Must have $= 0$. $27x^2+412x-436=0$ Must have integer coefficients. Accept a multiple of this.	A1 [6]
		Total 14 marks

Question	Scheme	Marks

Type MS · Finding an unknown constant from a condition on the roots

Answer 5 of 10 · Question 10 part a · 8 marks

2024 JUN · Paper 01 (regional R) · find part a in this mark scheme extract

10(a)	$\alpha+\beta=-\frac{k}{2}\qquad \alpha\beta=2$	B1
	$(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta,\quad \alpha^2-\beta^2=(\alpha+\beta)(\alpha-\beta)$	B1B1
	$\Rightarrow \alpha^2-\beta^2=(\alpha+\beta)\sqrt{(\alpha+\beta)^2-4\alpha\beta}$	
	$\Rightarrow \alpha^2-\beta^2=\left(-\frac{k}{2}\right)\sqrt{\left(-\frac{k}{2}\right)^2-4\times 2}=\frac{7\sqrt{17}}{4}$	M1
	$\Rightarrow \frac{k^4}{4}-8k^2-\frac{833}{4}=0\Rightarrow k^4-32k^2-833=0$ or any equivalent	M1A1
	$\Rightarrow (k^2-49)(k^2+17)=0\Rightarrow k=-7$ *	M1A1 cso [8]
(b)	Product: $\left[(\alpha-\beta)(\alpha+\beta)=\alpha^2-\beta^2=\frac{7\sqrt{17}}{4}\right]$ from (a)	B1
	Sum: $(\alpha-\beta)+(\alpha+\beta)=\frac{\sqrt{17}}{2}+\frac{7}{2}=\left(\frac{\sqrt{17}+7}{2}\right)$	M1
	$x^2-\left(\frac{\sqrt{17}+7}{2}\right)x+\frac{7\sqrt{17}}{4}=0\Rightarrow 4x^2-2\left(\sqrt{17}+7\right)x+7\sqrt{17}=0$	M1A1 [4]
		Total 12 marks

Type MS · Finding an unknown constant from a condition on the roots

Answer 6 of 10 · Question 7 part a.i · 0 marks

2022 JAN · Paper 02 · find part a.i in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Finding an unknown constant from a condition on the roots

Answer 7 of 10 · Question 7 part a.ii · 9 marks

2022 JAN · Paper 02 · find part a.ii in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Finding an unknown constant from a condition on the roots

Answer 8 of 10 · Question 8 part whole · 11 marks

2023 JAN · Paper 02 · find part whole in this mark scheme extract

8	$\alpha+\beta=4k\sqrt{2}$ and $\alpha\beta=2k^4-1$	B1 B1
	$(\alpha+\beta)^2=\alpha^2+\beta^2+2\alpha\beta\Rightarrow\left(4k\sqrt{2}\right)^2=66+2\left(2k^4-1\right)$	M1 A1
	$k^4-8k^2+16=0$	M1
	$\left(k^2-4\right)^2=0\Rightarrow k=2$	M1 A1
	$(\alpha+\beta)^3=\alpha^3+3\alpha^2\beta+3\alpha\beta^2+\beta^3$ **or** $\alpha^3+\beta^3=(\alpha+\beta)\left(\alpha^2+\beta^2-\alpha\beta\right)$	M1
	$\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$ **or** $\alpha^3+\beta^3=(\alpha+\beta)(66-\alpha\beta)$	A1
	$\alpha^3+\beta^3=\left(8\sqrt{2}\right)^3-3\times31\times8\sqrt{2}=280\sqrt{2}$ **or** $8\sqrt{2}(66-31)=280\sqrt{2}$	M1
	p=280	A1 (11)
		Total 11 marks

Type MS · Finding an unknown constant from a condition on the roots

Answer 9 of 10 · Question 7 part b · 4 marks

2025 NOV · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks
7(a)	$(\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2)=\alpha^3-\alpha^2\beta+\alpha\beta^2+\beta\alpha^2-\alpha\beta^2+\beta^3=\alpha^3+\beta^3$	B1 [1]
(b)	$\alpha+\beta=-4 \quad \alpha\beta=\frac{-k}{2}$	B1
	$\left[\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta\right]$	
	$\alpha^3+\beta^3=(\alpha+\beta)\left((\alpha+\beta)^2-2\alpha\beta-\alpha\beta\right)=(\alpha+\beta)\left((\alpha+\beta)^2-3\alpha\beta\right)$	M1
	$-94=(-4)\left((-4)^2-3\times\frac{-k}{2}\right)\Rightarrow\frac{47}{2}=16+\frac{3k}{2}\Rightarrow k=5$ *	M1A1 cso [4]
	ALT	
	$\alpha+\beta=-4 \quad \alpha\beta=-\frac{k}{2}$	[B1
	$\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$	M1
	$-94=(-4)^3-3\times-\frac{k}{2}\times-4\Rightarrow-30=-6k$	M1
	$k=5$ *	A1]
(c) (i)	$(\alpha-\beta)^2=\alpha^2-2\alpha\beta+\beta^2=(\alpha+\beta)^2-4\alpha\beta$	M1
	$\alpha-\beta=\sqrt{(-4)^2-4\times-\frac{5}{2}}=\sqrt{26}$ *	M1A1 cso [3]
(ii)	$\alpha^3-\beta^3=(\alpha-\beta)^3+3\alpha\beta(\alpha-\beta)$	M1
	$=\left('\sqrt{26}'\right)^3+3\left(\frac{-5}{2}\right)\left('\sqrt{26}'\right)$	
	$=\frac{37\sqrt{26}}{2}$	A1 [2]
		Total 10 marks

Type MS · Finding an unknown constant from a condition on the roots

Answer 9 of 10 · Question 7 part b · 4 marks

2025 NOV · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
7 (a)	B1	A full expansion with no errors.
(b)	B1	For the correct values of the product and sum
	M1	For the correct algebra $\alpha^3+\beta^3$ in a form that values for $\alpha+\beta$ and $\alpha\beta$ [in terms of a] can be substituted.
	M1	For substituting their values of $\alpha+\beta$ and $\alpha\beta$ into their expression for $\alpha^3+\beta^3$ and solving a **linear** equation to find the value of k.
	A1 cso	For the correct value of $k = 5$
	ALT	
	B1	For the correct values of the product and sum
	M1	For the correct algebra $\alpha^3+\beta^3$ in a form that values for $\alpha+\beta$ and $\alpha\beta$ [in terms of a] can be substituted using $\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$
	M1	For substituting their values of $\alpha+\beta$ and $\alpha\beta$ into their expression for $\alpha^3+\beta^3$ and solving a **linear** equation to find the value of k.
	A1 cso	For the correct value of $k = 5$
(c) (i)	M1	For the correct algebra for $(\alpha-\beta)^2$
	M1	For substituting the correct values for the sum and product and take the positive square root.
	A1 cso	For the correct value with no errors seen.
(ii)	M1	For the correct algebra to find $\alpha^3-\beta^3$ in a form such that the sum and product can be substituted in directly.
	A1	For the correct value of $\alpha^3-\beta^3$

Type MS · Finding an unknown constant from a condition on the roots

Answer 9 of 10 · Question 7 part b · 4 marks

2025 NOV · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Finding an unknown constant from a condition on the roots

Answer 10 of 10 · Question 6 part a · 5 marks

2023 JUN · Paper 02 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

6(a)	$\alpha+\beta=-\frac{5}{2}\quad \alpha\beta=-\frac{p}{2}$	B1
	$\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$	M1
	$\alpha^3+\beta^3=\left(-\frac{5}{2}\right)^3-3\left(-\frac{p}{2}\right)\left(-\frac{5}{2}\right)=-\frac{125}{8}-\frac{15p}{4}=-\frac{215}{8}$	M1M1
	$\Rightarrow p=3$	A1
	ALT	[5]
	$\alpha^3+\beta^3=(\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2)=(\alpha+\beta)\left((\alpha+\beta)^2-3\alpha\beta\right)$	[M1]
(b)	Sum:	
	$\frac{\alpha+\beta}{\alpha^2}+\frac{\alpha+\beta}{\beta^2}=\frac{\alpha\beta^2+\beta^3+\alpha^3+\alpha^2\beta}{\alpha^2\beta^2}=\frac{\alpha^3+\beta^3+\alpha\beta(\alpha+\beta)}{\alpha^2\beta^2}$	M1
	ALT	
	$\frac{\alpha+\beta}{\alpha^2}+\frac{\alpha+\beta}{\beta^2}=\frac{\alpha^2(\alpha+\beta)+\beta^2(\alpha+\beta)}{\alpha^2\beta^2}=\frac{(\alpha^2+\beta^2)(\alpha+\beta)}{\alpha^2\beta^2}$	M1
	$=\frac{\left((\alpha+\beta)^2-2\alpha\beta\right)(\alpha+\beta)}{\alpha^2\beta^2}$	
	$\frac{-\frac{215}{8}+\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)}{\left(-\frac{3}{2}\right)^2}=-\frac{185}{18}$ allow $\frac{185}{18}$ if they start with a negative.	B1
	Product:	
	$\left(\frac{\alpha+\beta}{\alpha^2}\right)\times\left(\frac{\alpha+\beta}{\beta^2}\right)=\frac{(\alpha+\beta)^2}{\alpha^2\beta^2}=\frac{\left(-\frac{5}{2}\right)^2}{\left(-\frac{3}{2}\right)^2}=\frac{25}{9}$	M1A1 [5]
	Equation:	
	$x^2+\frac{185}{18}x+\frac{25}{9}=0\Rightarrow 18x^2+185x+50=0$ oe	
		Total 10 marks

Type MS · Finding an unknown constant from a condition on the roots

Answer 10 of 10 · Question 6 part a · 5 marks

2023 JUN · Paper 02 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	B1	For the correct expression/values for **BOTH** the sum and product. This must be identified, or implied from their working.
	M1	For the correct algebra to find $\alpha^3+\beta^3$ in terms of $\alpha+\beta$ and $\alpha\beta$ • $\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$ • $\alpha^3+\beta^3=(\alpha+\beta)\left((\alpha+\beta)^2-3\alpha\beta\right)$ • Or any other algebra, but do not award this mark until the values of $\alpha+\beta$ and $\alpha\beta$ can be substituted in directly.
	M1	For substituting their values of the sum and product into their expression for $\alpha^3+\beta^3$
	M1	For solving the **linear** equation in p Allow one slip in their working.
	A1	For the correct value of p with no errors.
(b)	M1	For the correct algebra for the sum of roots. This must be such that the given value of $\alpha^3+\beta^3$, with their values of $\alpha\beta$ and $(\alpha+\beta)$ can be substituted in. If they use the **ALT** they will not need $\alpha^3+\beta^3$ Some candidates will reverse the sign at this stage in anticipation of the reversal required in the equation.
	M1	For substituting in the given value for $\alpha^3+\beta^3$, and their values for $\alpha+\beta$ and $\alpha\beta$ into their expansion for the sum. If they use the **ALT** they will not need $\alpha^3+\beta^3$
	B1ft	For the correct value of the product of roots. You must follow through their p. You do not need to check the calculation once you see the correct algebra with a correct substitution.
	M1	For forming an equation with their sum and product. Ft the sign of their sum. Some candidates reverse the sign when finding the sum. Watch out for that! Accept this without = 0
	A1	For a correct equation including = 0 There is no follow through on this mark. **NB** accept a correct equivalent equation provided it has integer coefficients. e.g. $36x^2+370x+100=0$
	SC – they solve the equation and find values for α and β Award marks for correct work seen above in part (b) • If they do not obtain an expansion for the sum into which $\alpha^3+\beta^3$, $\alpha\beta$ and $(\alpha+\beta)$ cannot be directly substituted – first M0 • If they do not substitute $\alpha^3+\beta^3$, $\alpha\beta$ and $(\alpha+\beta)$ but other values based on α and β - second M0 • If they cannot substitute $\alpha\beta$ and $(\alpha+\beta)$ into the product – B0 Then allow marks for forming the equation as above.	

Type MS · Finding an unknown constant from a condition on the roots

Answer 10 of 10 · Question 6 part a · 5 marks

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Question	Scheme	Marks